

SST-MAE: Supplementary Material

Table S1: Sample sizes and class distributions for the four SNPs. The 80/20 genotype-level split yields 11,376 training and 2,880 test plant images. Heterozygotes are included in total counts but excluded from binary class columns.

SNP	Samples		Class Distribution			
	Training (n)	Test (n)	Class 0		Class 2	
			train	test	train	test
ant.loc5	11,376	2,880	10,152	1,656	1,224	1,224
ant.loc9	11,376	2,880	10,512	2,376	792	432
pale.loc4	11,376	2,880	4,824	1,368	6,192	1,512
serr.loc5	11,376	2,880	9,360	2,088	2,016	792

Table S2: Few-shot training set sizes (10–100%) per SNP. Fractions are based on the total homozygous training pool. The fixed test set (held-out genotypes) remains fully labeled.

SNP	Training Fraction					Fixed test set
	10%	30%	50%	70%	100%	
ant.loc5	1,138	3,413	5,688	7,963	11,376	2,880
ant.loc9	1,130	3,391	5,652	7,913	11,304	2,808
pale.loc4	1,102	3,305	5,508	7,711	11,016	2,880
serr.loc5	1,138	3,413	5,688	7,963	11,376	2,880

Table S3: Computational cost summary for SST-MAE pre-training and inference. The computing node running AlmaLinux 9.8 is equipped with 8 NVIDIA RTX A6000 GPUs (48 GB VRAM each), 1.1 TB of RAM, and 50 TB of high-performance storage. Model pre-training was performed on a single GPU node.

Metric	Value
GPU	NVIDIA RTX A6000 (48 GB)
Training epochs	100 (early stopping at 72)
Batch size	16
Total parameters	4.2 million
Encoder parameters	2.8 million
Decoder parameters	1.4 million
Pre-training time	$\approx$ 8.5 hours
Inference (feature extraction) per plant	$\approx$ 45 ms

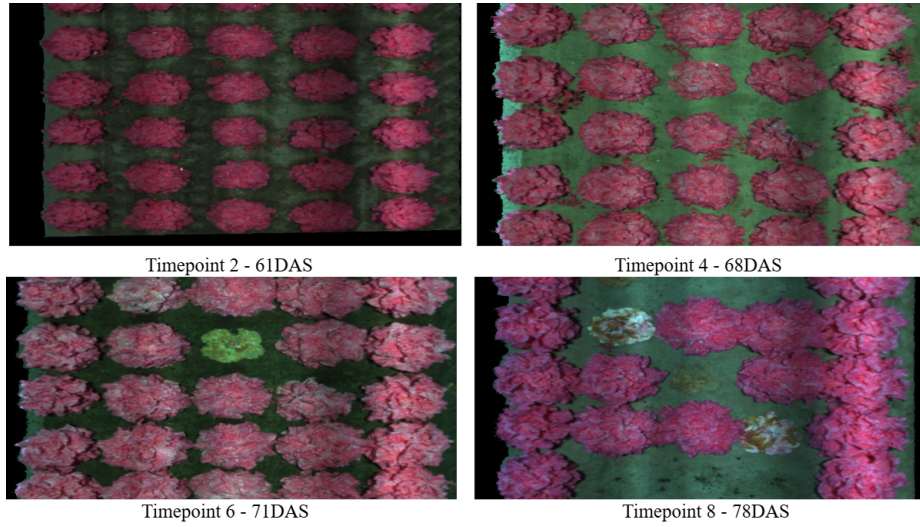


Fig.S1: Representative hyperspectral RGB composite images from the lettuce panel at selected developmental time points (DAS is the days after sowing).